



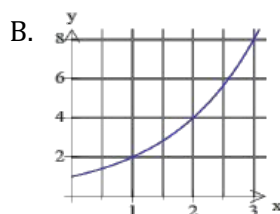
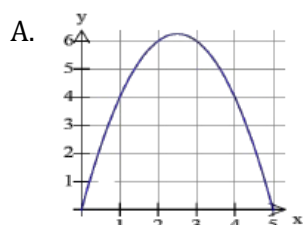
Non-Linear Relationships

Topic Test 1 Solution

Total Marks: 30

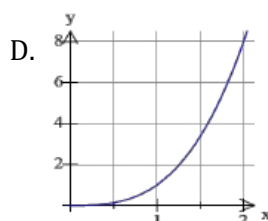
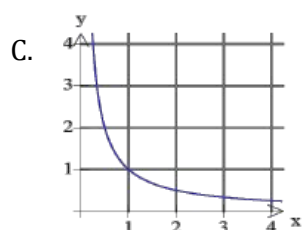
Time: 50 minutes

- 1 Which graph best represents the function $y = \frac{1}{x}$? 1

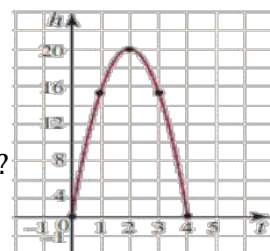


Hyperbola

C



- 2 A ball is thrown in the air. Its height h metres, after t second can be given by the formula $h = 20t - 5t^2$. The graph of the function is shown below.



- a. What is the height of the ball 3 seconds after it is thrown in the air? 1
b. What limitations are there on this model? $h = 15$ m 1

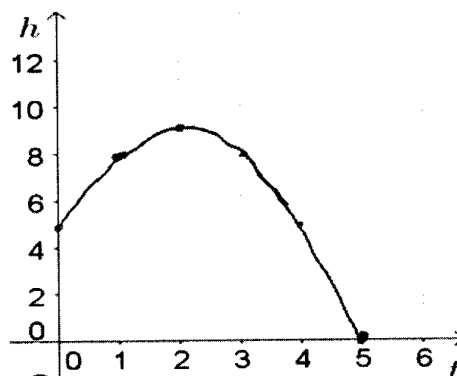
If $t < 0$ or $t > 4$ the value of h becomes negative which is not possible given the ball cannot go below ground level.

- 3 An object is thrown in the air. It takes 5 seconds for it to reach the ground. The height of the object is given by the formula: $h = -t^2 + 4t + 5$ where h is the height and t is time in seconds.

- a. Complete the following table of values.

t	0	1	2	3	4	5
h	5	8	9	8	5	0

- b. Draw the graph of $h = -t^2 + 4t + 5$ on the number plane below.



- c. What is the maximum height reached by the object? 9 1
d. When is the maximum height reached? $t = 2 \text{ sec}$ 1
e. What was the initial height of the object? 5 1

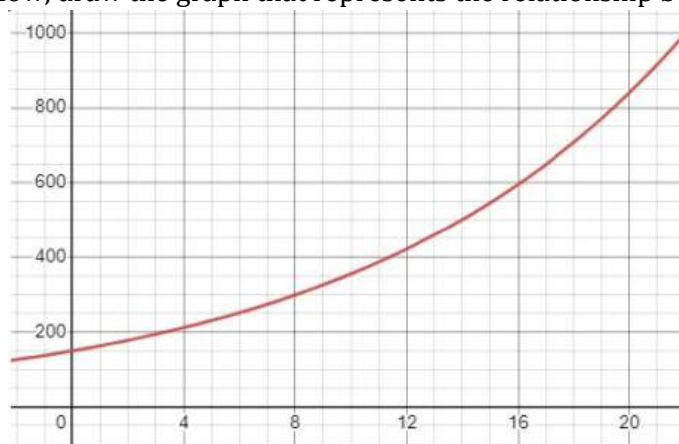
- 4 The number of cases of infection of a certain disease grows exponentially according to the function $c = 150(1.09)^d$, where c is the number of cases and d is the time in days.

Number of days (d)	0	4	8	12	16	20
Number of cases (c)	150	212	299	422	596	

- a. How many cases are there after 20 days? Answer correct to the nearest whole number? 1

$$\begin{aligned}
 \text{When } d &= 20 \\
 c &= 150(1.09)^{20} \\
 &= 840.6616 \dots \\
 &\approx 841 \text{ cases}
 \end{aligned}$$

- b. On the grid below, draw the graph that represents the relationship between d and c . 2



- c. Using the graph drawn, estimate the time taken, in days, for the number of cases to reach 800. Give your answer correct to the nearest whole number. 1

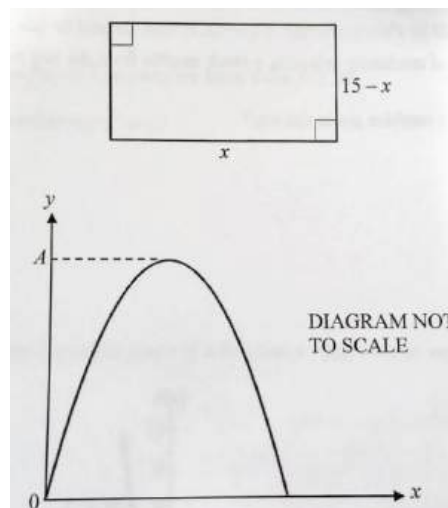
Approximately 19 days

- 5 To determine the maximum area (A) of a rectangle with a perimeter of 30m, Melanie sketched the following diagrams in her notebook. 1

The point A is the turning point on the parabolic graph.

Melanie correctly determined the maximum area of the rectangle (in square metres) by finding the value of A.

What is the value?



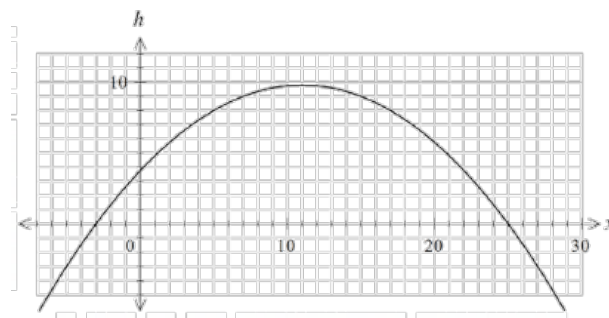
$$\begin{aligned}
 &x(15-x) \\
 &= -x^2 + 15x \\
 &= -(7.5)^2 + 15(7.5) \\
 &56.25
 \end{aligned}$$

- 6 A ball is thrown off the top of a building, and lands on the ground. The path of the ball can be modelled by the equation

$$h = 3.75 + 1.1x - 0.05x^2$$

where

- h is the height above ground level in metres.
- x is the horizontal distance from the starting position in metres.



- What is the initial height of the ball when it is thrown? **3.75 m** 1
- Find the exact height of the ball when its height is at a maximum. 2

The maximum is when $x = \frac{-3+25}{2} = 11$
 $h = 3.75 + 1.1(11) - 0.05(11)^2 = 9.8$

1

- Give, with reasons, one range of x values for which this model will not be practical.

$x < 0$ is not practical because it starts at
 $x = 0$

OR

$x > 25$ is not practical because then the
height would be below 0.

- 7 The frequency (F) (number of vibrations per second) measured in Hertz (Hz), of a string on a harp, varies inversely with the square root of the length of the string (L) measured in cm. A string of length 25 cm on a harp has frequency of 10 Hz.

- Show that the situation can be modelled by the equation: $F = \frac{50}{\sqrt{L}}$ 1

$$F = \frac{50}{\sqrt{L}} \quad 10 = \frac{50}{\sqrt{25}} \quad 10 = \frac{50}{5}$$

2

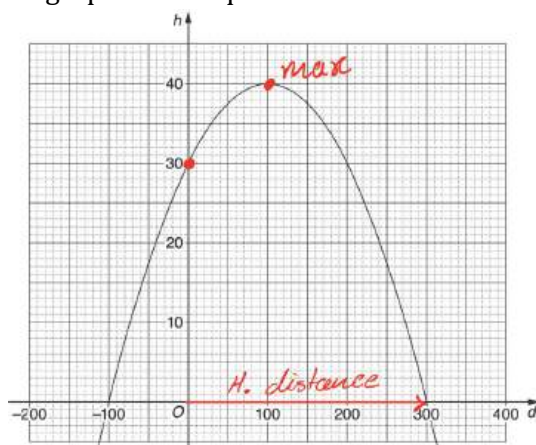
- What length of string would produce a frequency of 12.5 Hz on the harp?

$$12.5 = \frac{50}{\sqrt{L}} \\ \sqrt{L} = \frac{50}{12.5} \\ \sqrt{L} = 4 \quad L = 16 \text{ cm}$$

- 8 If pressure (P) varies inversely with volume (V), write a formula to correctly express in terms of P and V . 2

$$P = \frac{k}{V}$$

- 9 A ball is thrown upward from a raised platform. The path of the ball's trajectory is modelled by $h = -0.001d^2 + 0.2d + 30$ where h is the height above the ground and d is the horizontal distance the ball travels. A graph of the equation is shown below.



- From what height is the ball thrown? **30m** 1
- What is the maximum distance above the ground the ball reaches? **40m** 1
- What is the horizontal distance travelled? **300m** 1
- What is the height when $d = 215.2$? 1

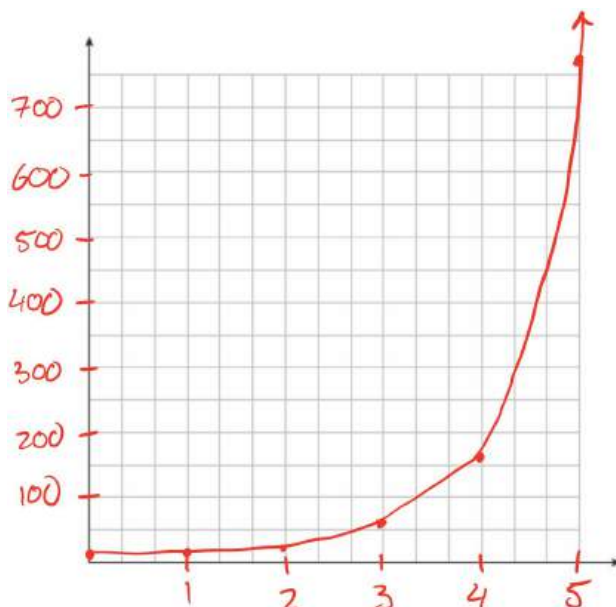
$$h = -0.001(215.2)^2 + 0.2(215.2) + 30 = 26.7m$$

- 10 An infectious disease grows exponentially according to the equation $I = 3.8^t$, where I is the number of infections after t days.

- Create a table of values of I and t (express the number of infections to the nearest whole number). 2

t	0	1	2	3	4	5
I	1	4	14	55	209	792

- Using part i), graph the exponential function on the axis below. 2





Non-Linear Relationships

Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

- 1 What is the minimum value of $y = 2x^2 - 10x + 7$? 1

- A. -10
B. -5.5
C. -5
D. 2.5

x	1	2	3	4
y	-1	-5	-5	-1

$x = 2.5, y = -5.5$

- 2 The time taken to build a StrongBuild house is inversely proportional to the number of people working on the task. A team of six people can build the house in 30 days. The schedule requires that the house be built in 18 days. How many people are required to have the house built in 18 days? 3

$$\begin{aligned} T &= \frac{k}{n} \\ 30 &= \frac{k}{6} \\ \therefore k &= 180 \\ T &= \frac{180}{n} \quad T = 18, \\ 18 &= \frac{180}{n} \\ n &= \frac{180}{18} = 10 \text{ people} \end{aligned}$$

- 3 The thickness T in centimetres of a tree trunk grows according to the function $T = 16(1.11)^n$ where n is the time in years after the first measurement.

- a. What is the dependent variable in the function? T 1
- b. Calculate the thickness of the tree trunk after 6 years. Give your answer correct to the nearest centimetre. $16(1.11)^6 \approx 30 \text{ cm}$ 2

- 4 The volume of a gas at constant temperature varies inversely to the pressure. What is the effect on the volume of a gas when the pressure is tripled? 1

- A. The volume stays the same. $V = \frac{k}{P}$, where V is volume and P is pressure.
- B. The volume is divided by 3. Let $P = 3$ and constant $k = 3$.
- C. The volume is tripled. $V = \frac{3}{3} = 1$
- D. The volume is cubed. If P is tripled, this gives:

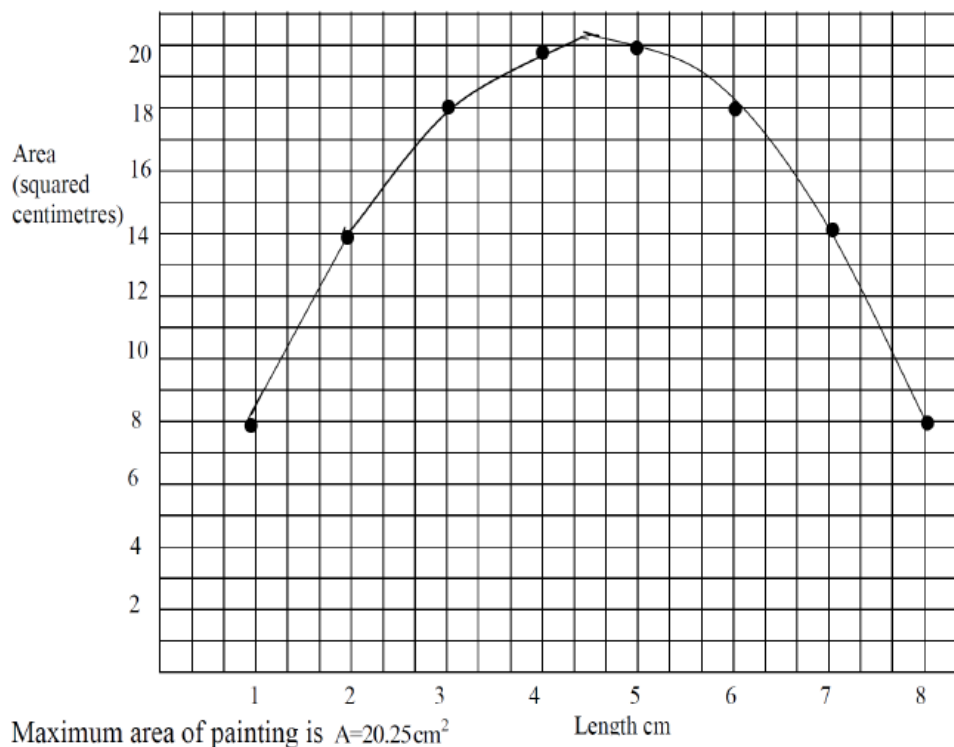
$$V = \frac{3}{9} = \frac{1}{3}$$

Therefore, V is one-third of the original volume (that is, the original volume is divided by 3).

- 5 a. A rectangular painting has a perimeter of 18 cm. Complete the following table. Units are in cm. 1

Length	1	2	3	4	5	6	7	8
Width	8	7	6	5	4	3	2	1
Area	8	14	18	20	20	18	14	8

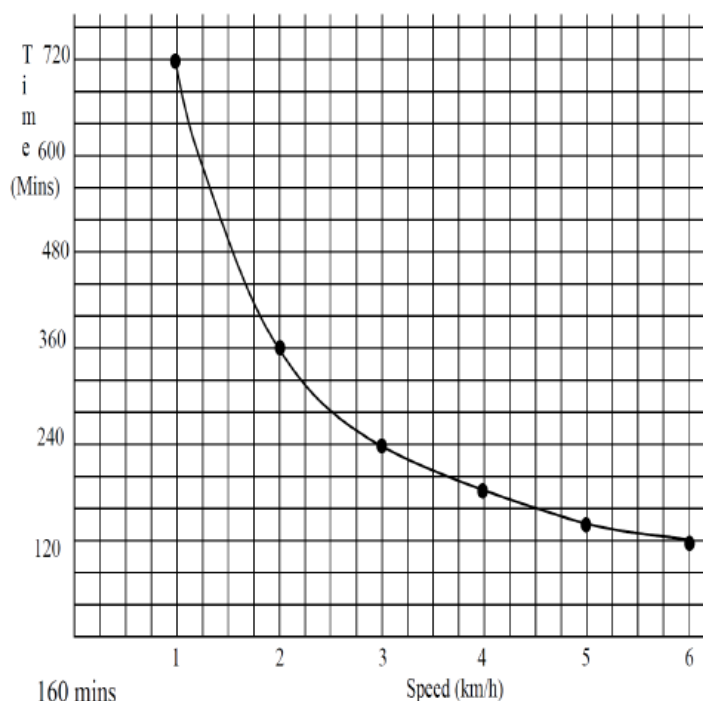
- b. On the following graph paper, graph length versus area and hence estimate the maximum area of the painting. Use the horizontal axis to represent length. 3



- 6 a. The time taken to walk from point A to point B is given by $T = \frac{720}{s}$, where T is in minutes and s is in km/h. Complete the following table. 1

s	1	2	3	4	5	6
T	720	360	240	180	144	120

- b. Graph T for the values of s from 1 to 6 and hence state the time taken to travel from A to B at a speed of 4.5 km/h. Let the vertical axis be time. 3



- 7 A company makes and sells fibre optic cable. It can sell, at most, 200 kilometres of cable in a week. The profit can be modelled by $P = 275x - x^2 - 2000$ where x is the number of kilometres of fibre optic cable, $x \leq 200$ and P is the profit in dollars.

- a. Use the profit equation, P , to find how much money the company loses if it does not sell any cable. 1

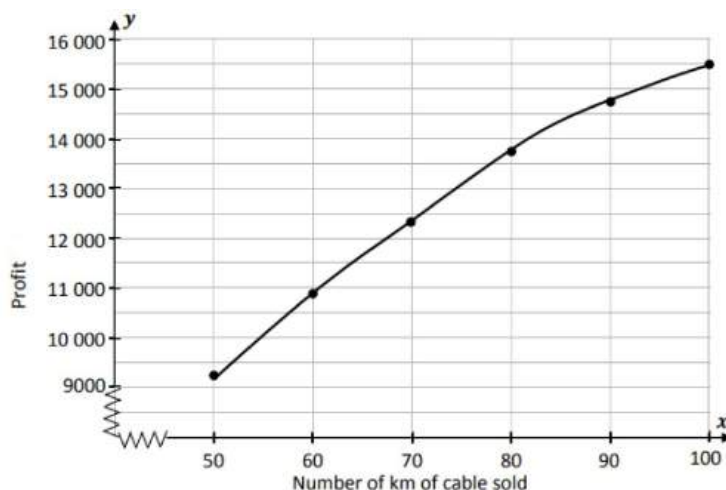
$$P = 275(0) - (0)^2 - 2000 = -2000$$

The company loses \$2 000 if it does not sell any cable.

- b. The table below shows some of the data representing the profit on sales. Use the profit equation, P , to complete the table. 1

Number of km of cable sold (x)	50	60	70	80	90	100
Profit (\$)	9 250	10 900	12 350	13 600	14 650	15 500

- c. Use the data in the table to draw the graph of the profit equation on the axes below. 2



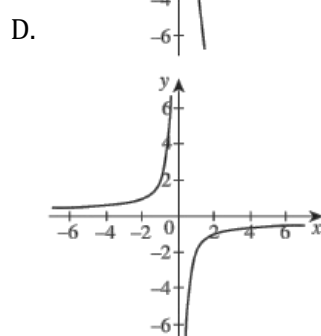
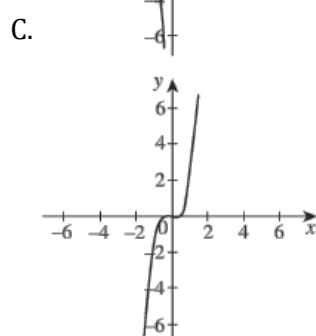
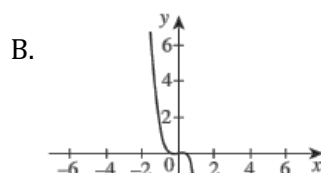
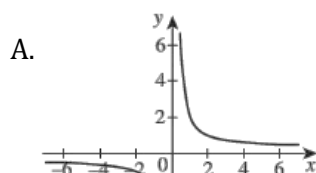
- d. Use your graph to estimate the lower and upper range of sales (in km of cable) in order to make a profit of between \$10 000 and \$14 000 in a particular week. Show your work on the graph above. 1

Lower Value = 55

Upper Value = 83

(Lines to be shown on diagram)

- 8 Which of the following could represent the graph of the function $y = \frac{2}{x}$? 1

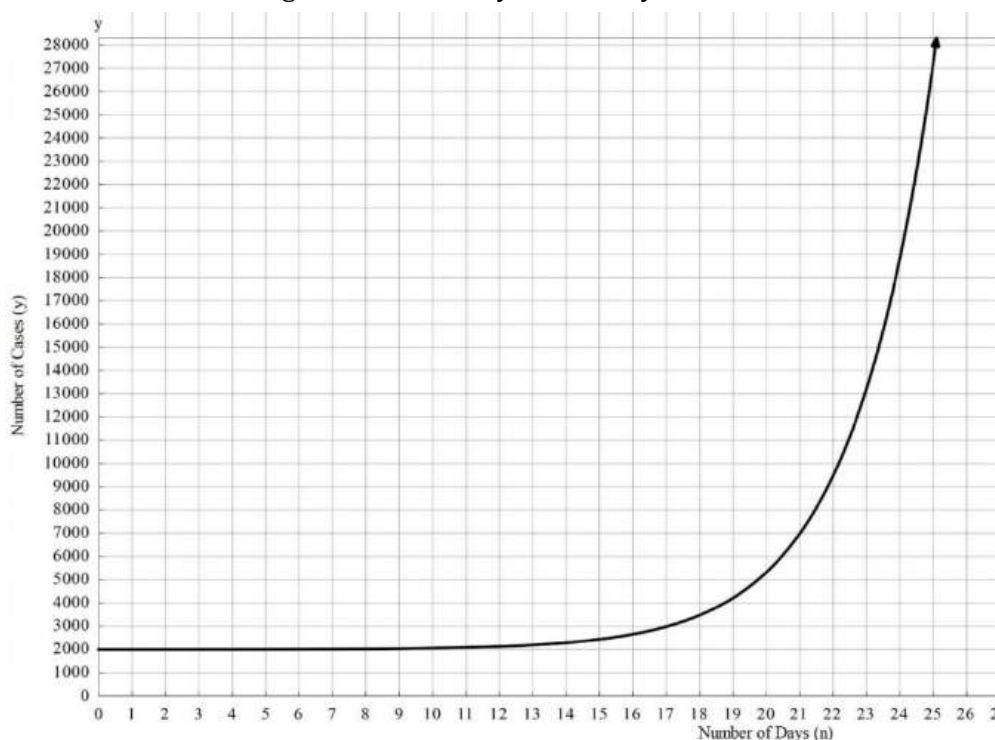


A

- 9 The graph shows the number of coronavirus cases, y , at time n days. Initially (when $n = 0$) the number of cases is 2000.

- a. Find how much the cases grew between day 17 and day 23.

2



Day 17 = 3 000 cases Day 23 = 13 000 cases
Growth = 13 000 – 3 000 = 10 000 cases

- b. For the given example, the number of cases can be modelled by the equation.

1

$y = 1.5n + 2000$. Explain why this mathematical model could differ from what happens in real life.

Governments may implement restrictions meaning the growth rate of 1.5 cases per day will reduce/growth rate will slow down.

- 10 The distance, d , in kilometres that a speed camera can see when placed at a height, h , in metres

above the ground is given by the formula $d = \sqrt{\frac{h}{6.5}}$.

- a. If the camera is placed 4 metres above ground, how far can it see? Give your answer correct to two decimal places.

1

$$d = \sqrt{\frac{4}{6.5}} = 0.7844 \dots \approx 0.78 \text{ km}$$

- b. Rearrange the formula $d = \sqrt{\frac{h}{6.5}}$ to make h the subject.

1

$$d = \sqrt{\frac{h}{6.5}}$$

$$d^2 = \frac{h}{6.5}$$

$$h = 6.5d^2$$

- 11 A disease's growth rate refers to how quickly the number of infections changes each day. The number of cases of infection of a certain disease grows exponentially according to the function $c = 220(1.07)^d$, where c is the number of cases and d is the time in days.

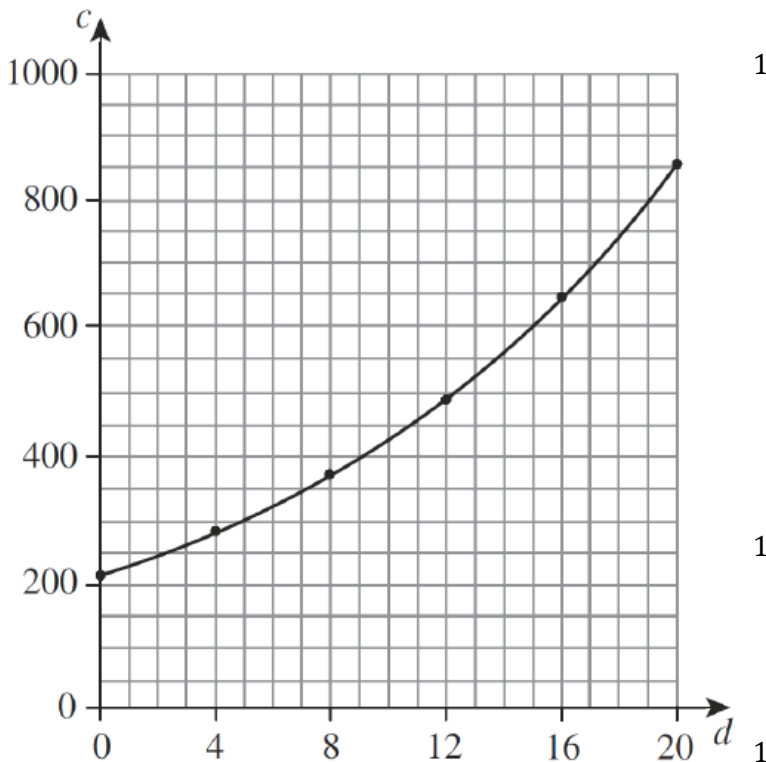
Number of days (d)	0	4	8	12	16	20
Number of cases (c)	220	288	378	495	649	?

- a. What is the number of cases after 20 days? Give your answer correct to the nearest whole number.

When $d = 20$:

$$\begin{aligned}
 c &= 220(1.07)^{20} \\
 &= 851.3305 \dots \\
 &\approx 851 \text{ cases}
 \end{aligned}$$

- b. On the grid below, draw the graph that represents the relationship between d and c .
- c. Using the graph drawn, or otherwise, estimate the time taken, in days, for the number



Reading the graph obtained in part (b) gives approximately 19 days.

of cases to reach 800. Give your answer correct to the nearest whole number.